LIFTING OF SPECIALIZATIONS AND OF GENERALIZATIONS IS STABLE UNDER COMPOSITION OF CONTINUOUS MAPS

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Allowed standard prerequisites (editorial).

- The specialization relation between points and the meaning of the phrases specializations lift and generalizations lift along a continuous map. Set up in the author's section on specialization, immediately before the counted statement. Fixes the two properties whose composition-stability the statement asserts.

Lemma 0.1. *Suppose $f : X \to Y$ and $g : Y \to Z$ are continuous maps of topological spaces. If specializations lift along both f and g then specializations lift along $g \circ f$. Similarly for "generalizations lift along".*

Proof. Let $z' \rightsquigarrow z$ be a specialization in Z and let $x' \in X$ such as $g \circ f(x') = z'$. Then because specializations lift along g, there exists a specialization $f(x') \rightsquigarrow y$ of $f(x')$ in Y such that $g(y) = z$. Likewise, because specializations lift along f, there exists a specialization $x' \rightsquigarrow x$ of x' in X such that $f(x) = y$. It provides a specialization $x' \rightsquigarrow x$ of x' in X such that $g \circ f(x) = z$. In other words, specialization lift along $g \circ f$. □

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